

WASP-11b

R-0731

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_240130 · created 2026-07-28T00:59:56+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460339.6784 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.011
Transit depth (Rp/Rs)^2	2.15 +/- 0.31 [%]
Orbital Inclination (inc)	88.1 +/- 0.83
Airmass coefficient 1 (a1)	0.992 +/- 0.011
Airmass coefficient 2 (a2)	0.006 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#3 - [483, 324]
Optimal Aperture	3.8
Optimal Annulus	11.2
Transit Duration (day)	0.1002 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.147 $\pm$ 0.011 vs 0.131 (deviation 1.27 σ)
Duration fit vs published	0.1002 d vs 0.1075 d (deviation 0.91 σ)
Mid-time O-C	-7.6 min
Depth SNR	11.7
Residual scatter	1.12 %

Light curve

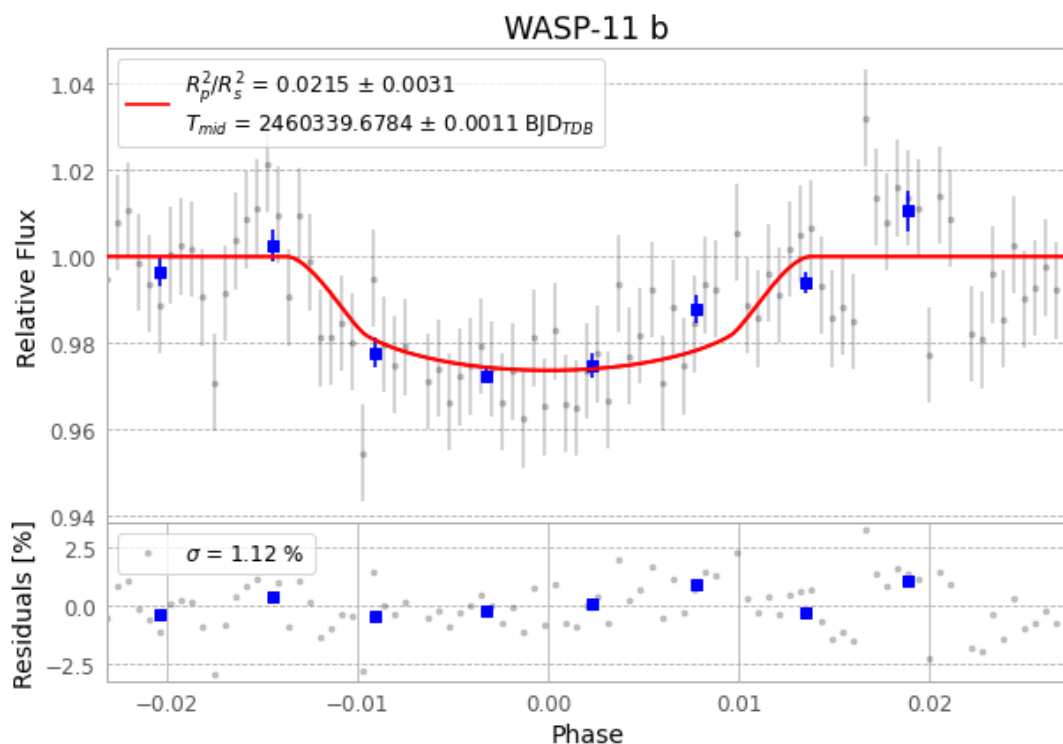


Figure 1 — FinalLightCurve_WASP-11 b_2024-01-29.png (SHA-256 284efeb8237f3e72...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 281], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6145758791930e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

91 FITS frames (60 MB), HATP-10240130020915.FITS → HATP-10240130063915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-240130.log (faeb71c96113...), FinalLightCurve_WASP-11 b_2024-01-29.png (284efeb8237f...), FinalLightCurve_WASP-11 b_2024-01-29.csv (269512c9f461...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:59Z.